NCoder 732 FOC Magnetic Encoder

Technical Specification & Design Reference

Variant Documentation

Variant A: JST-GH Bottom (SMD)
Variant B: 1.25mm Pitch (THT)

Status: Production Ready
Application: Aerospace, Drone Payloads, and Advanced Robotics

Generated for Engineering Manufacturing Review

Contents

1 System Overview

The NCoder 732 is a highly integrated, industrial-grade absolute magnetic angle encoder tailored for Field-Oriented Control (FOC) motor driver systems. Designed for extreme-reliability environments, the board utilizes signal multiplexing to transmit both high-speed SPI (absolute position) and ABZ (incremental quadrature) data over a unified 6-pin interface.

This document encompasses the technical specifications for two distinct mechanical variants of the board. Both variants share identical logical architectures and component configurations, differing exclusively in their connector mounting strategies to accommodate varying robotic harness requirements.

2 Mechanical Variants

2.1 Variant A: JST-GH Bottom (SMD)

This variant utilizes a surface-mount JST-GH connector (J2) designed for low-profile integration. The connector is mounted on the bottom side of the PCB, allowing the top side (housing the MA732 sensor) to sit flush against the motor housing with minimal air gap.

- **Connector Type:** 6-Pin JST-GH (SM06B-GHS-TB or equivalent).
- **Primary IC Designators:** U1 (Sensor), S1 (Multiplexer), U2 (TVS Array).

2.2 Variant B: 1.25mm Pitch (THT)

This variant employs a through-hole technology (THT) 1.25mm pitch pin header (J4). This configuration is highly robust against mechanical sheer forces and is suitable for custom wire-harness soldering or standard breadboard prototyping during the early stages of FOC driver development.

- **Connector Type:** 6-Pin 1.25mm Pitch THT Header.
- **Primary IC Designators:** U3 (Sensor), S2 (Multiplexer), U4 (TVS Array).

3 Bill of Materials (BOM) Synthesis

The following table outlines the critical components for both variants, cross-referencing their specific designators.

4 Hardware Configuration Logic

The functional output of the board is controlled entirely by hardware via the `SEL_NET` (Variant A) or `SEL_NET1` (Variant B) line. This state is manipulated using the 3-pad solder jumper (J1 / J3).

Variant A Desig.	Variant B Desig.	Component	Primary Function
U1	U3	MA732GQ-Z	Main Absolute Magnetic Angle Sensor (QFN-16).
S1	S2	TS5A23157DGSR	Dual SPDT Analog Switch (Multiplexer).
U2	U4	TPD4E004DRYR	4-Channel TVS Diode Array (ESD Protection).
J2	J4	6-Pin Header	External interface to the upstream FOC Controller.
J1	J3	Solder Jumper	3-Pad hardware mode selection (SPI vs. ABZ).
R1 (1kΩ)	R10 (1kΩ)	Resistor	Z-Pulse collision protection.
R4 (10kΩ)	R12 (10kΩ)	Resistor	SEL_NET pull-up (Defaults to SPI Mode).
R3, R5-R7	R11, R13-R15	33Ω Resistor	Series termination for transmission line damping.

Mode Selection Matrix

Default State (SPI Mode):
The 10kΩ pull-up resistor (R4 / R12) holds the network at +3.3V. The S1/S2 multiplexer routes the MOSI and MISO lines to the connector. *Requires no solder on the jumper.*

Hardwired State (ABZ Mode):
Bridging the center pad of the jumper to the GND pad forces the network LOW, overpowering the pull-up resistor. The multiplexer switches to route Sensor_A and Sensor_B lines to the connector.

Critical Warning:
Bridging all three pads on the jumper creates a dead short between +3.3V and GND, which will cascade failure to the main FOC driver voltage regulator.

5 Signal Integrity & Interface Protection Network

To ensure aerospace-grade reliability, the data lines bridging the encoder to the motor controller are protected by a two-stage impedance and transient network.

5.1 ESD Clamping (Stage 1)

As raw signals enter the connector, they terminate directly onto the pads of the TPD4E004DRYR TVS array. This component is rated to safely shunt massive electrostatic discharges to the local ground plane before they puncture the sensitive gate oxides of the multiplexer or MA732 silicon.

5.2 Transmission Line Damping (Stage 2)

Following the TVS clamp, signals immediately pass through 33Ω series damping resistors (R3, R5, R6, R7 for Variant A). Fast digital clock edges (SCLK) over long drone harnesses inherently produce transmission line reflections. These resistors act in conjunction with trace capacitance to filter and smooth the leading edges, eliminating bit-read errors.

5.3 The CLK/Z Shared Topology

To conserve connector pins, Pin 5 acts as a bidirectional channel depending on the selected mode:

- **Incoming SPI Clock:** Travels through the TVS diode, through the 33Ω series resistor, and terminates at the `SCLK` pin.
- **Outgoing ABZ Index:** Pulses out of the `Z` pin, passes through the 1kΩ collision protection resistor (R1/R10), joins the main track, and exits via the connector. This 1kΩ resistor prevents catastrophic IC failure in the event of bidirectional signal contention.

6 PCB Layout & Routing Specifications

The physical routing of the copper directly dictates the efficacy of the protection network.

1. **Flow-Through Routing:** T-junctions are strictly prohibited at the TVS array interface. Traces must enter the TVS pad and exit from the identical pad geometry to strictly enforce the ESD clamping path.
2. **Component Sequencing:** As defined by the internal layout rule: *"TVS array sits within 5 mm of the Connector. Series R sits near the source IC. Order from source: IC → 33Ω → trace → TVS → Connector Pin"*.
3. **Sensor Centering:** The MA732 must be positioned at the absolute geometric center of the PCB boundaries to ensure perfect diametric alignment with the motor shaft's magnet.
4. **Ground Plane:** The bottom layer must consist of a solid, unbroken copper Ground plane to act as an electromagnetic noise shield against the inductive switching fields of the FOC motor.